Referee package – Proposition A: the isotropic Gaussian-Hartree layer is the strict diagonal minimum, and the off-diagonal reduces to the additive-energy floor (class-wide H-LAYER closure)

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This referee document concerns Proposition A (the Prop-A folder of B2-PROPA-HLAYER): that the isotropic Gaussian-Hartree layer G_* is the strict comparison minimum, with the H-LAYER residuals closed class-wide on the primary shell-supported competitor class $\mathcal{A}_{\mathrm{adm}}$ ($T' \leq 13$). **Scope:** $\mathcal{A}_{\mathrm{adm}}$ (registered crystalline shell / shell-union, $T' \leq 13$) at the operating point. **Not claimed:** the full $\mathcal{C}_{\mathrm{full}}$ comfortable extension is the Reading-H package; here the margin on $\mathcal{A}_{\mathrm{adm}}$ is the comfortable $T' \leq 13$ one.

2. The result

The T-016..T-024 programme closes every ANALYTIC H-LAYER residual class-wide on $\mathcal{A}_{\mathrm{adm}}$: (D) the diagonal Hessian at G_* is block-diagonal in angular momentum with $l \geq 1$ curvature $\frac{1}{2}G_*^{-2} \geq \frac{1}{2}r_R^2 > 0$ and $l = 0$ adding $\frac{3}{2}u_{\mathrm{eff}} > 0$; (O) the off-diagonal reduces to the additive-energy floor with $R_{\mathrm{lead}}(13) = 0.650 < 1$; (S) the third-cumulant joint $= \times 1.097 > 1$ with the analytic threshold $T' < 60.4$.

3. Structure of the argument

(D) An arbitrary perturbation δG is decomposed in spherical harmonics; the rotation-invariant interaction curves only $l = 0$, so the Hessian is block-diagonal and strictly positive in every sector. The strict diagonal-*minimum* conclusion invokes the global/convex component of T-016; the Hessian computation alone gives local strict stability. (O) The condensate block is diagonal in the channel index (a sum of independent channel terms), so its operator norm is the worst single-channel ratio $R_{\mathrm{lead}} = 23.2(1 + K_{\mathrm{floor}})I$, bounded by the floor pin $K_{\mathrm{floor}} \leq T' \leq 13$. (S) MARGIN cancels in the certified joint, leaving a function of ρ_{lat} with joint $> 1 \iff T' < 60.4$; since $T' \leq 13 < 60.4$ on $\mathcal{A}_{\mathrm{adm}}$, the joint evaluates to joint $= \times 1.097 > 1$.

4. Numerical reproduction

The nine reproduction scripts (each with self-test asserts, each exiting 0) reproduce the class-wide second-cumulant stability, the Gershgorin row-sums, the diagonal convexity, the off-diagonal additive-energy floor and the constant certificate. The full manifest (no wildcard) is:

```
9 reproduction scripts (codes/vacuum/; each asserts, exits 0):
  (D) diagonal isotropy / convexity:
        hdiag_convexity_probe.py
        res5_016_isotropy_infimum_core.py
  (O) off-diagonal additive-energy floor + operator norm:
```

```
        hdiag_offdiag_additive_energy.py
        hdiag_offdiag_constant_certificate.py
        res1_hdiag_offdiag_floor.py
        res5_019_exchange_scalar_identification.py
        hdiag_gershgorin_rowsum.py
  (S) third-cumulant selection (class-wide):
        res5_020_classwide_secondcumulant_stability.py
  route audit:
        res5_029_t7_route_audit.py
```

Run them from the bundle and diff against `expected/`. *Supersession note:* `hdiag_gershgorin_rowsum.py` is retained as a row-sum / additive-energy support script; its older "exchange sign not fixed by $u_{\text{eff}} > 0$" framing is superseded by T-019 (`res5_019`), which reframes the TECT local interaction as having no A-independent attractive Fock exchange and a stabilising mean-field off-diagonal Hessian $f''(M) = \frac{3}{2}u_{\text{eff}} > 0$. The live route uses $R_{\text{lead}} < 1$ and the T-019 reframing; the audit `res5_029` records this supersession.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.

(β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is PUBLISHED-BUNDLE CONFIRMED within the stated $\mathcal{A}_{\text{adm}}$ scope; the claim-ledger tier remains T6 and does not enact the full C_{full} Reading-H theorem.

(γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

```
Result ID:              B2-PROPA-HLAYER / Prop-A-T6-260611
Precise statement:      On the primary shell-supported class A_adm
                        (T'<=13), the T-016..T-024 programme closes
                        every analytic H-LAYER residual: (D)
                        block-diagonal isotropy Hessian (entropy floor
                        (1/2)r_R^2>0, breathing (3/2)u_eff>0); (O)
                        off-diagonal = additive-energy floor with
                        R_lead(13)=0.650<1; (S) third-cumulant
                        joint=x1.097>1 (analytic threshold T'<60.4).
Scope:                  A_adm (registered crystalline
                        shell/shell-union, T'<=13), operating point.
                        Comfortable margin.
Evidence grade:         class-wide analytic closure on A_adm (T6/T7);
                        EXECUTED (9 scripts PASS).
Expected output:        R_lead(13)=0.650<1; joint=x1.097>1;
                        T'_threshold=60.4; all 9 scripts PASS.
Reproduction command:   run the 9 scripts listed in Sec.4
                        (codes/vacuum/; each exits 0); full manifest
                        in Sec.4, no wildcard.
```

Falsifier:	an A_adm competitor with R_lead>=1 or joint<=1; or a perturbation breaking the block-diagonal positivity.
Tier before / after:	A_adm closure, T6 (main-line supporting). PUBLISHED-BUNDLE CONFIRMED (operator 2026-06-11): Prop-A-T6-260611.
No-overclaim statement:	class-wide closure on A_adm (T'<=13, comfortable); the full C_full extension is the Reading-H package.
Next required action:	none for this package (PUBLISHED as Prop-A-T6-260611); the full C_full closure is the separate Reading-H package.
Publication:	confirmed after reconstructed clean-run from the uploaded bundle: 9/9 scripts, 44/44 asserts PASS. Scope A_adm, T'<=13; does NOT independently enact full C_full Reading-H.